

Financial Mathematics

(MATH35400/MATHM5400)

Problem Class 1 – 23th January 2026 ¹

1. (2020 exam, Q1)

(a) Explain in one sentence each:

- i. the difference between an option and a forward contract,
- ii. the difference between a call option and a put option,
- iii. the difference between a European option and an American option.

(b) The current price for AKOCOM shares is £37.50 per share. A put option on 1 share of AKOCOM maturing in exactly half a year with a strike price of £30 is traded at £11. The price of the corresponding call option with same strike price and same maturity is currently £18.50. Assume that the interest rate for lending money as well as depositing money on a bank account is 5% per annum.

- i. Use the put-call parity to show that there exist arbitrage opportunities in the model above.
- ii. Describe explicitly what actions an arbitrageur would carry out today and in six months time to attain riskless profit. You can assume that the arbitrageur does not have to pay any fees and that short selling is possible.
[Hint: You might need to distinguish two cases to describe the actions in six months time.]

2. (2021 exam, Q1 (similar)) Stocks of telecommunication company *PlinTo* are currently traded at $S_0 = £8$ per share. A financial company wants to price a European call option on *PlinTo* shares with strike price $K = £5$ maturing in exactly one year. They have access to a bank account with an annual interest rate $r = 0.125$.

Alex suggests to model the stock by assuming that the stock price in a year's time S_1 is either £11 with probability 0.6 or £3 with probability 0.4. Alex suggests to work out the expected value of the call option and use that as the time 0 price for the option.

- (a) What is the value of the European call option at time 1 in each case?
- (b) What price would Alex charge?
- (c) Find a replicating portfolio for the European call option, i.e. find $H = (H_0, H_1)$ such that the value of the European call option at time 1 is identical to the portfolio where one invests H_0 on the bank account and buys H_1 stocks of

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PlinTo. Then determine its time 0 value C_0 and compare with the price that Alex calculated in i.

- (d) What would an arbitrageur do at time 0 if the company would sell the option for Alex' price?